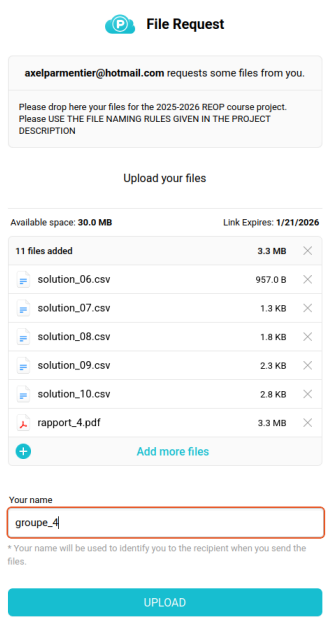
Operations research project

Projects should be done by groups of three students, and submitted on the following pcloud link (https://e.pcloud.com/#/puplink?code=57n7ZyOVvWrjDCymfMzVll7naOYIzw9Fy) before January 20th, 23:59. For each group, please upload the following five files in a single upload on the link above.

- A report with the `pdf` format named `rapport_GROUP.pdf`. This report must :
  - contain the group number and the name of all the students of the group,
  - be at most *4 pages* with a font size of at least 11 point (*-2 points by additional page*)
  - As large language models are authorized for code generation, the report should be especially clear in its presentation of the algorithm used and its analysis.
- The solution files which should be named `solution_01.csv`, `solution_02.csv`, `solution_03.csv`, `solution_04.csv`, `solution_05.csv`, `solution_06.csv`, `solution_07.csv`, `solution_08.csv`, `solution_09.csv`, `solution_10.csv` containing the solution of the instances provided.

In file names, `GROUP` should be replaced by your group number. In the form on the pcloud link, please put your group number in the field "Your name".

For instance, if you are in group 4, your submission on pcloud should look like this.



Please respect the nomenclature, the `json` and `pdf` formats, and the single upload per group (*-1 point for each rule not respected*).

1 Input data

Vehicles Califrais can rent from an unlimited fleet of vehicles. The vehicles are split into F families. Inside each vehicle family f, the vehicles are interchangeable and have common features:

- A maximum capacity w_f^{capacity}.
- A daily rental cost c_f^{rental}.

- A unit fuel cost per meter traveled c_f^{fuel}.
- A unit radius cost per squared meter c_f^{radius} (see below).
- A speed s_f
- A parking time p_f
- Temporal variation coefficients α_f, β_f, for computing travel times (see below).

Orders Califrais must serve a set of orders denoted by $\mathcal{I}$. Each order $i \in \mathcal{I}$ is defined by:

- Its ID i.
- Its spatial coordinates (φ_i, λ_i), where φ_i is the latitude and λ_i is the longitude (both in degrees).
- Its weight $w_i \in \mathbb{R}_+$.
- Its delivery time window $[t_i^{\min}, t_i^{\max}]$.
- Its delivery duration ℓ_i.

The Califrais depot, or warehouse, is denoted by $i = 0$. All trucks depart from there at $t^{\text{dep}} = 0$.

Distances Califrais operates on the Paris region road network, but taking into account every street would be too complex. Instead, we work on a complete directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ linking all orders $i \in \mathcal{I}$ to each other and to the depot 0. In other words, its vertices and edges are given by

$$\mathcal{V} = \mathcal{I} \cup \{0\} \qquad \text{and} \qquad \mathcal{E} = \{(i,j) \in \mathcal{V}^2 : i \neq j\}$$

Given two nodes i and j, we will make use of the following distances:

- Manhattan distance $\delta^{\text{M}}(i,j) = |x_j - x_i| + |y_j - y_i|$
- Euclidean distance $\delta^{\text{E}}(i,j) = \sqrt{|x_j - x_i|^2 + |y_j - y_i|^2}$

To perform the conversion from spherical coordinates expressed in degrees to geodesic segments with a length in meters, we use the correspondence between angles in radians and arc lengths:

$$\begin{aligned} y_j - y_i &= \rho \frac{2\pi}{360}(\varphi_j - \varphi_i) \\ x_j - x_i &= \rho \left(\cos \frac{2\pi}{360}\varphi_0\right) \frac{2\pi}{360}(\lambda_j - \lambda_i) \end{aligned}$$

where $\rho = 6.371 \times 10^6$ m is the Earth's radius and φ_0 is the latitude of the depot (arbitrarily).

Travel times While physical distances are constant, travel times can depend on the vehicle type (big vehicles are slower) and the time of day (rush hour is bad on the *périphérique*). We express the travel time $\tau_f(i, j \mid t)$ between two nodes as a reference travel time multiplied by a time-dependent factor:

$$\tau_f(i, j \mid t) = \tau_f(i,j) \cdot \gamma_f(t)$$

The reference travel time is an affine function of the Manhattan distance:

$$\tau_f(i,j) = \frac{\delta^{\text{M}}(i,j)}{s_f} + p_f$$

where s_f is the speed of the vehicle and p_f is the parking time. The influence of time is periodic with period $T = 1$ day $= 86400$ s, so we decompose it as a truncated Fourier series with angular frequency $\omega = 2\pi/T$:

$$\gamma_f(t) = \sum_{n=0}^{3} (\alpha_f(n)\cos(n\omega t) + \beta_f(n)\sin(n\omega t))$$

You can assume that these travel times satisfy the First In, First Out (FIFO) property: if you leave later, you arrive later.

2 Solutions

Routes A solution to the delivery scheduling problem is expressed as a set $\mathcal{R}$ of routes. Each route $r \in \mathcal{R}$ is defined by the following attributes. The three first ones are enough to define the routes as decision variables, while the last two are auxiliary and can be derived from the first three and the problem data.

- The family of vehicle used $f_r \in \{1, \dots, F\}$.
- The number of orders served n_r.
- The sequence of vertices visited, starting and ending at the depot and visiting orders in between:

$$(i_r^0, i_r^1, \dots, i_r^{n_r}, i_r^{n_r+1}) \quad \text{with} \quad i_r^0 = i_r^{n_r+1} = 0 \quad \text{and} \quad i_r^k \in \mathcal{I} \text{ for } k \in \{1, \dots, n_r\}.$$

- The sequence of arrival times at vertices $(a_r^1, \dots, a_r^{n_r}, a_r^{n_r+1})$ (not defined for the initial vertex).
- The sequence of departure times from vertices $(d_r^0, d_r^1, \dots, d_r^0, d_r^{n_r})$ (not defined for the final vertex).

We say that an order i belongs to a route r, and we write $i \in r$, if there exists $k \in \{1, \dots, n_r\}$ such that $i_r^k = v$. We say that an edge $e = (i, j)$ belongs to a route r, and we write $(i, j) \in r$, if there exists $k \in \{1, \dots, n_r\}$ such that $i_r^{k-1} = i$ and $i_r^k = j$.

Constraints We now enumerate all the constraints that must be satisfied by a solution $\mathcal{R}$:

1. Each order belongs to exactly one route (so the routes are disjoint):

$$i \in \bigsqcup_{r \in \mathcal{R}} r \qquad \forall i \in \mathcal{I},$$

   where $\bigsqcup$ denotes the disjoint union.

2. Each vehicle is loaded with the total weight of all the orders on its route. This load must respect its maximum capacity:

$$\sum_{i \in r} w_i \leq w_{f_r}^{\text{capacity}} \qquad \forall r \in \mathcal{R}$$

3. Times:

   (a) A vehicle can only depart from an order after having arrived and then performed the delivery:

$$d_r^k \geq a_r^k + \ell_{i_r^k}$$

   (b) A vehicle can only arrive at its first vertex after having departed and traveled from the depot:

$$a_r^1 \geq d_r^0 + \tau_{f_r}(0, i_r^1 \mid t^{\text{dep}}) \qquad \forall r \in \mathcal{R}$$

(c) A vehicle can only arrive at a subsequent vertex after having departed and traveled from the previous vertex:

$$a_r^{k+1} \geq d_r^k + \tau_{f_r}(i_r^k, i_r^{k+1} \mid d_r^k) \qquad \forall r \in \mathcal{R}$$

(d) Arrival times at each order must fall within the order's delivery window:

$$t_{i_r^k}^{\min} \leq a_r^k \leq t_{i_r^k}^{\max} \qquad \forall r \in \mathcal{R}, \forall k \in \{1, \ldots, n_r\}$$

Note that parking at an order's location and waiting until the window opens is permitted (strictly speaking, the arrival time is the time at which delivery starts).

Objective Califrais strives to minimize the total cost of all routes, composed of several components:

$$\min_{\mathcal{R}} c(\mathcal{R}) = \sum_{r \in \mathcal{R}} \left(c^{\text{rental}}(r) + c^{\text{fuel}}(r) + c^{\text{radius}}(r) \right)$$

Individual route costs are defined as follows:

1. The daily rental cost, which pushes towards cheaper vehicle types:

$$c^{\text{rental}}(r) = c_{f_r}^{\text{rental}}$$

2. The total fuel cost based on the Manhattan distance traveled, which pushes towards fewer or shorter routes:

$$c^{\text{fuel}}(r) = c_{f_r}^{\text{fuel}} \sum_{k=0}^{n_r} \delta^{\text{M}}(i_r^k, i_r^{k+1})$$

3. A penalty for the squared Euclidean radius of the route (excluding the depot), weighted by c^{radius}, which pushes towards clustered deliveries:

$$c^{\text{radius}}(r) = c_{f_r}^{\text{radius}} \left| \frac{1}{2} \max_{\substack{k \in \{1,\ldots,n_r\} \\ \ell \in \{1,\ldots,n_r\} \\ k \neq \ell}} \delta^{\text{E}}(i_r^k, i_r^\ell) \right|$$

3 Data format

Conventions and units

- Times are expressed in seconds, taking midnight as a reference for times (5:20 AM is given as 19200 for instance).
- Distances are expressed in meters.
- Weights and capacities are expressed in kilograms.
- Longitude and latitude are expressed in degrees.
- Costs are expressed in euros.

Input files We now describe the data provided by Califrais, as a set of two CSV files.

`vehicles.csv` contains one row per vehicle family (F in total), with the following columns:

- `family` : f
- `max_capacity` (kg): w_f^{capacity}
- `rental_cost` (€): c_f^{rental}
- `fuel_cost` (€/m): c_f^{fuel}
- `radius_cost` (€/m^2): c_f^{radius}
- `speed` (m/s): s_f
- `parking_time` (s): p_f
- `fourier_cos_0` (unitless): $\alpha_f(0)$
- `fourier_cos_1` (unitless): $\alpha_f(1)$
- `fourier_cos_2` (unitless): $\alpha_f(2)$
- `fourier_cos_3` (unitless): $\alpha_f(3)$
- `fourier_sin_0` (unitless): $\beta_f(0)$
- `fourier_sin_1` (unitless): $\beta_f(1)$
- `fourier_sin_2` (unitless): $\beta_f(2)$
- `fourier_sin_3` (unitless): $\beta_f(3)$

`instance.csv` contains one row for the depot and one row per order ($|\mathcal{V}|$ in total), with the following columns:

- `id` : i
- `latitude` (degrees): φ_i
- `longitude` (degrees): λ_i
- `order_weight` (kg): w_i (empty for the depot)
- `window_start` (s): $t_i^{\min}$ (empty for the depot)
- `window_end` (s): $t_i^{\max}$ (empty for the depot)
- `delivery_duration` (s): ℓ_i (empty for the depot)

For the depot, the last 4 columns are empty.

Note that the `vehicles.csv` is the same for all problem instances, while the `instance.csv` varies.

Solution file The solution you provide must consist of a single CSV file:

`routes.csv` contains one row per route ($|\mathcal{R}|$ in total) with the following columns:

- `family` f_r
- `order_1` i_r^1
- `order_2` i_r^2
- ...
- `order_`n_r $i_r^{n_r}$

To build this file, first find the maximum number of orders served by a route in your solution $N = \max_{r \in \mathcal{R}} n_r$. The resulting table should have $2+N$ columns, and the routes shorter than N should end with empty columns (not filled with a space or a null placeholder of any kind).

Thanks to the problem constraints, the arrival and departure times at all vertices can be reverse-engineered from the sequence of orders, so they do not need to be specified.

4 Instructions

4.1 Questions (6 points)

Answer the following questions in your report. These questions focus on the mathematical modeling of the problem. For these questions, unless specified otherwise, assume a simplified version of the problem with a homogeneous fleet (single vehicle family with capacity Q), constant travel times, no rental cost, and no radius cost.

1. **3-index formulation (MILP):**
   Propose a Mixed-Integer Linear Programming formulation using 3-index binary variables x_{ij}^k (equal to 1 if vehicle k travels from i to j). Include constraints for flow conservation, capacity, and time windows.

2. **2-index formulation (MILP):**
   Propose a formulation using 2-index variables x_{ij} (equal to 1 if a vehicle travels from i to j), assuming a homogeneous fleet. Use Miller-Tucker-Zemlin (MTZ) like constraints to eliminate subtours and enforce capacity.

3. **Dynamic Programming (Single Route):**
   Consider the problem of finding the optimal route for a single vehicle visiting a specific subset of customers $S \subseteq \mathcal{I}$. Propose a Dynamic Programming approach to solve this Traveling Salesman Problem with Time Windows (TSPTW). Define the state space and the recurrence relation.

4. **Lower Bound :**
   We wish to estimate a lower bound on the minimum number of vehicles required, regardless of the distance traveled and costs. Propose such a lower bound and give an algorithm to compute it.

4.2 Algorithm (14 points)

Using a strategy of your choice, propose a solution for the 10 instances of growing size provided.

- You will turn in a file in the desired format giving your solution for the instance provided.
- The quality of the resolution strategies used and their presentation in the report will be scored out of 8 points.
  - Quality of the approach provided (*4 points*).
    - Prove any interesting results on these approaches (accuracy, complexity, etc.).
  - Rigor and quality of writing will be given special consideration. (*4 points*)
    - The algorithms must be provided in a readable way (https://en.wikipedia.org/wiki/Pseudocode)
    - For the numerical results, provide the costs of the solutions, their optimality / gap.
    - The numerical results must be presented in a readable way (table, captioned figure).
- The quality of the solutions provided will be noted on 6 points.
  - The score of a team is the sum of the cost of the solutions returned for each instance (the large instances therefore have a much stronger weight). The teams will be ranked by score (the team with the lowest score will have the best score).

    On the 6 points, 4 will be linked to the ranking:
    - 4 points + 2 bonus points for the first group
    - 4 points + 1 bonus point for the second
    - 4 points for the third group

    - 3 points for those who are in the first quarter
    - 2 points for those who are in the second quarter
    - 1 point for those who are in the third quarter
    - 0 for the others
  - Your solutions will be verified with the verification code provided to you. Please email me at axel.parmentier@enpc.fr if you identify an error in the subject or in the verification code.

The problem is difficult to solve.

- **Don't start the project at the last minute.** Ask the teachers in your groups questions.
- Feel free to simplify / decompose it to get feasible solutions.
- If you wish, you can use solvers to solve the problem. However, since the instance is large, it is unlikely that a solver can find a good solution if applied to the whole instance. This does not prevent you from using a solver for a sub-problem.

5 Resources

5.1 Language

You should use the language you are the most comfortable with.

- See e.g. https://gdalle.github.io/phd-resources/tutorials/python/ for more resources mathematics with `python`.
- See e.g. https://gdalle.github.io/IntroJulia/ for more resources on mathematics with `julia`.

5.2 MILP solvers

You can for instance use the open-source MILP solvers `HiGHS` or `SCIP`. If you wish, you can also use a commercial solver such as `Gurobi` with a free academic license. See e.g.

- https://www.cvxpy.org/tutorial/advanced/index.html#mixed-integer-programs on how to use a MILP solver in `python`.
- https://jump.dev/JuMP.jl/stable/ on how to use a MILP solver in `julia`.

6 Acknowledgements

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